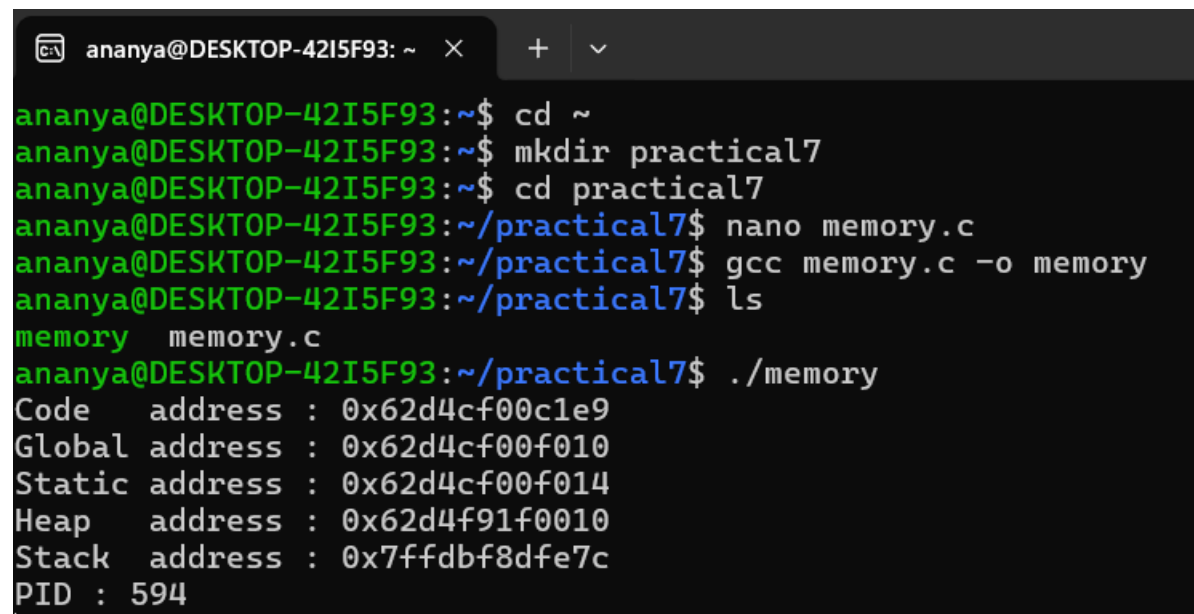


OS-PRACTICAL

Practical-7

7. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.



```
ananya@DESKTOP-42I5F93: ~  
ananya@DESKTOP-42I5F93:~$ cd ~  
ananya@DESKTOP-42I5F93:~$ mkdir practical7  
ananya@DESKTOP-42I5F93:~$ cd practical7  
ananya@DESKTOP-42I5F93:~/practical7$ nano memory.c  
ananya@DESKTOP-42I5F93:~/practical7$ gcc memory.c -o memory  
ananya@DESKTOP-42I5F93:~/practical7$ ls  
memory  memory.c  
ananya@DESKTOP-42I5F93:~/practical7$ ./memory  
Code    address : 0x62d4cf00c1e9  
Global  address : 0x62d4cf00f010  
Static  address : 0x62d4cf00f014  
Heap    address : 0x62d4f91f0010  
Stack   address : 0x7ffdbf8dfe7c  
PID : 594
```

```

ananya@DESKTOP-42ISF93: ~ $ cd ~/practical7
ananya@DESKTOP-42ISF93: ~/practical7$ cat /proc/594/maps
62d4cf00b000-62d4cf00c000 r--p 00000000 08:30 40601
62d4cf00c000-62d4cf00d000 r-xp 00001000 08:30 40601
62d4cf00d000-62d4cf00e000 r--p 00002000 08:30 40601
62d4cf00e000-62d4cf00f000 r--p 00002000 08:30 40601
62d4cf00f000-62d4cf010000 rw-p 00003000 08:30 40601
62d4f91f0000-62d4f9211000 rw-p 00000000 00:00 0
763564800000-763564828000 r--p 00000000 08:30 13628
763564828000-7635649c0000 r-xp 00028000 08:30 13628
7635649c0000-763564a0e000 r--p 001c0000 08:30 13628
763564a0e000-763564a12000 r--p 0020d000 08:30 13628
763564a12000-763564a18000 rw-p 00211000 08:30 13628
763564a14000-763564a21000 rw-p 00000000 00:00 0
763564bd2000-763564bd5000 rw-p 00000000 00:00 0
763564bdc000-763564bde000 rw-p 00000000 00:00 0
763564bde000-763564be2000 r--p 00000000 00:00 0
763564be2000-763564be4000 r--p 00000000 00:00 0
763564be4000-763564be6000 r-xp 00000000 00:00 0
763564be6000-763564be7000 r--p 00000000 08:30 13625
763564be7000-763564c16000 r-xp 00001000 08:30 13625
763564c16000-763564c21000 r--p 00030000 08:30 13625
763564c21000-763564c23000 r--p 0003b000 08:30 13625
763564c23000-763564c24000 rw-p 0003d000 08:30 13625
763564c24000-763564c25000 rw-p 00000000 00:00 0
7ffdbf8c1000-7ffdbf8e2000 rw-p 00000000 00:00 0
ananya@DESKTOP-42ISF93: ~/practical7$ cat /proc/594/maps | grep heap
62d4f91f0000-62d4f9211000 rw-p 00000000 00:00 0
ananya@DESKTOP-42ISF93: ~/practical7$ cat /proc/594/maps | grep stack
7ffdbf8c1000-7ffdbf8e2000 rw-p 00000000 00:00 0

```

```
ananya@DESKTOP-42I5F93:~/practical7$ cat /proc/594/status
```

```
Name:      memory
Umask:     0022
State:     S (sleeping)
Tgid:      594
Ngid:      0
Pid:       594
PPid:      324
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 1000
NStgid:    594
NSpid:     594
NSpgid:    594
NSsid:     324
Kthread:   0
VmPeak:    2768 kB
VmSize:    2768 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     1812 kB
VmRSS:     1812 kB
RssAnon:           100 kB
RssFile:           1712 kB
RssShmem:          0 kB
VmData:    224 kB
VmStk:     132 kB
VmExe:      4 kB
VmLib:     1828 kB
VmPTE:      48 kB
VmSwap:     0 kB
HugetlbPages:      0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:     1
SigQ:       1/14991
SigPnd:     0000000000000000
ShdPnd:     0000000000000000
```

